

E-Commerce Sales Dashboard

Complete Project Documentation — For Local Backup & GitHub Upload

Generated: July 28, 2026

Supabase Project: cykbfaxaejuizbfbxirr

Stack: React 18 + TypeScript + Vite + Supabase + Recharts

Read-Only Dashboard

No Auth

Single-Tenant

Dark Theme

RLS Enabled

1. Project Overview

This is a **read-only e-commerce analytics dashboard**. It visualizes sales data stored in a Supabase (PostgreSQL) database using interactive charts. There is no login or sign-up — the dashboard is designed as a public demo that anyone with the link can view.

- **Name:** ecommerce-sales-dashboard
- **Purpose:** Display revenue trends, top categories, customer segments, and seasonal trends
- **Owner tag on screen:** "Curated by Dana Analyst"
- **Port:** 3000 (dev server)

2. Tech Stack

Layer	Technology	Version
Frontend	React	18.3
Language	TypeScript	5.5
Build Tool	Vite	8.1
Charts	Recharts	2.12
Database	Supabase (PostgreSQL)	—
Font	Inter (Google Fonts)	400-800

3. Environment Variables (.env)

These are stored in the `.env` file at the project root:

```
VITE_SUPABASE_URL=https://cykbfaaejuizbfbxirr.supabase.co
VITE_SUPABASE_ANON_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc3MiOiJzdXBhYmFzZSIsInJlZiI6ImN5aWMyYXhhZCw01aXoiZmJ4a
```

Note: The anon key is safe for frontend use — it only allows reading data. There is no service role key in this project. Never commit a service role key to GitHub.

4. Database Details

Supabase Project

- **URL:** <https://cykbfaxaejuizbfbxirr.supabase.co>
- **Project ID:** cykbfaxaejuizbfbxirr
- **Schema:** public

Tables Overview

Table	Rows	Purpose	RLS
categories	6	Product categories (name, slug)	Enabled
products	24	Products with price, cost, category link	Enabled
customers	200	Customer records (name, email, first order date)	Enabled
orders	3,383	Order headers (customer, date, total, status)	Enabled
order_items	6,767	Line items per order (product, quantity, unit price)	Enabled

Table Schemas

categories

```
id      uuid PRIMARY KEY DEFAULT gen_random_uuid()
name    text NOT NULL
slug    text UNIQUE
```

products

```
id          uuid PRIMARY KEY DEFAULT gen_random_uuid()
name        text NOT NULL
category_id uuid NOT NULL → categories.id (ON DELETE CASCADE)
price       numeric(10,2) NOT NULL
cost        numeric(10,2) NOT NULL DEFAULT 0
```

customers

```
id          uuid PRIMARY KEY DEFAULT gen_random_uuid()
name        text NOT NULL
email       text UNIQUE NOT NULL
first_order_date date (nullable)
```

orders

```
id          uuid PRIMARY KEY DEFAULT gen_random_uuid()
customer_id uuid NOT NULL → customers.id (ON DELETE CASCADE)
order_date  timestampz NOT NULL DEFAULT now()
total       numeric(12,2) NOT NULL DEFAULT 0
status      text NOT NULL DEFAULT 'completed'
```

order_items

```
id          uuid PRIMARY KEY DEFAULT gen_random_uuid()
order_id    uuid NOT NULL → orders.id (ON DELETE CASCADE)
product_id  uuid NOT NULL → products.id (ON DELETE CASCADE)
quantity    integer NOT NULL DEFAULT 1
unit_price  numeric(10,2) NOT NULL
```

Relationships

```
categories (1) → (many) products
products   (1) → (many) order_items
customers  (1) → (many) orders
orders     (1) → (many) order_items
```

Indexes

Index Name	Table	Column(s)
idx_orders_customer_id	orders	customer_id
idx_orders_order_date	orders	order_date
idx_products_category_id	products	category_id
idx_order_items_order_id	order_items	order_id
idx_order_items_product_id	order_items	product_id

RLS Policies (Current — Read-Only)

Each table has exactly **one SELECT policy**. No INSERT, UPDATE, or DELETE policies exist.

```
-- Example for each table (replace <table> with the table name):
CREATE POLICY "anon_select_<table>" ON <table> FOR SELECT
  TO anon, authenticated USING (true);
```

Tables and their SELECT policies:

- `anon_select_categories` on categories
- `anon_select_products` on products
- `anon_select_customers` on customers
- `anon_select_orders` on orders
- `anon_select_order_items` on order_items

Migrations Applied (3 total)

#	Filename	Description
1	20260727235701_create_ecommerce_schema.sql	Created all 5 tables, indexes, and initial policies
2	20260728001353_fix_rls_policies_split_per_verb.sql	Split FOR ALL policies into per-verb (SELECT/INSERT/UPDATE/DELETE)
3	20260728001914_remove_write_policies_readonly_dashboard.sql	Removed all write policies (read-only dashboard)

Edge Functions

None deployed.

5. File Structure

```
project/
├── .env                ← Supabase URL + anon key
├── .gitkeep
├── index.html          ← HTML entry, loads Inter font
├── package.json        ← Dependencies & scripts
├── package-lock.json
├── tsconfig.json       ← TypeScript config
├── tsconfig.tsbuildinfo
├── vite.config.ts      ← Vite config (port 3000)
├── src/
│   ├── main.tsx        ← React entry point
│   ├── App.tsx         ← Dashboard UI (KPIs + 4 charts)
│   ├── index.css       ← Dark theme styling
│   └── vite-env.d.ts
├── &boxboxh;─ lib/
│   ├── supabase.ts     ← Supabase client singleton
│   ├── types.ts        ← TypeScript interfaces
│   └── &boxboxh;─ queries.ts    ← All data-fetching functions
&boxboxh;─ supabase/
│   └── migrations/
│       ├── 20260727235701_create_ecommerce_schema.sql
│       ├── 20260728001353_fix_rls_policies_split_per_verb.sql
│       └── 20260728001914_remove_write_policies_readonly_dashboard.sql
```

6. Dashboard Features

KPI Cards (top row, 5 cards)

TOTAL REVENUE

All-time gross sales — sum of all order totals

TOTAL ORDERS

Count of all completed orders

AVG ORDER VALUE

Total revenue divided by total orders

TOTAL CUSTOMERS

Count of unique buyers

REPEAT CUSTOMER RATE

Percentage of customers with 2 or more orders

Charts (4 total)

1. Revenue Over Time

[Area Chart](#)

Monthly revenue trend across all categories. Blue gradient fill.

2. Top Categories by Revenue

[Horizontal Bar Chart](#)

Which product categories drive the most sales. Multi-colored bars.

3. Customer Segments

[Donut / Pie Chart](#)

One-time (1 order) vs Repeat (2-5 orders) vs Loyal (6+ orders).

4. Seasonal Trends

[Bar Chart](#)

Average revenue by calendar month. November & December highlighted in amber as peak months.

Data Queries (in queries.ts)

Function	What It Does
<code>fetchKpiSummary()</code>	Loads orders + customers, computes all 5 KPIs client-side
<code>fetchMonthlyRevenue()</code>	Groups orders by YYYY-MM for the area chart
<code>fetchCategoryRevenue()</code>	Joins order_items → products → categories for bar chart
<code>fetchCustomerSegments()</code>	Buckets customers by order count (1 / 2-5 / 6+)
<code>fetchSeasonalTrend()</code>	Averages revenue per calendar month across all years

7. Design / Theme

Element	Color / Value
Background	Deep navy gradient (#060d1f → #0a1535)
Card background	#0c1830
Card border	#1a2e5c
Primary (blue)	#2563eb
Accent (green)	#10b981
Warning / peak (amber)	#f59e0b
Error (red)	#ef4444
Text	#eff6ff
Muted text	#7da2d6
Font	Inter (400, 500, 600, 700, 800)
Border radius	14px

Responsive Breakpoints

- **Desktop:** 5-column KPI grid, 2-column chart grid
- **Tablet (≤1024px):** 2-column KPI grid, 1-column chart grid
- **Mobile (≤640px):** 1-column KPI grid, smaller headings

Interactions

- KPI cards lift up 2px and glow blue on hover
- Chart cards border highlights on hover
- Charts animate in over 800ms

8. How to Run Locally

```
# Install dependencies
npm install

# Start dev server (runs on port 3000)
npm run dev

# Production build
npm run build

# Preview production build
npm run preview
```

9. Security Notes

- App is **read-only** — no write operations anywhere in the code
- RLS enabled on all 5 tables; only SELECT policies exist
- No authentication / login — designed as a public demo dashboard
- Anon key is safe for frontend (read-only access by design)
- **Never** commit a service role key to GitHub
- The `.env` file contains only the anon key (safe to commit for this demo, but for production apps consider adding `.env` to `.gitignore`)

10. GitHub Upload Guide

Step 1: Create a .gitignore file

Create a file named `.gitignore` in the project root with this content:

```
node_modules/  
dist/  
.env  
*.tsbuildinfo
```

Important: If you add `.env` to `.gitignore`, anyone who clones your repo will need to create their own `.env` file with the Supabase URL and anon key. For this demo project, the anon key is safe to commit. If you prefer to keep it private, use `.gitignore`.

Step 2: Initialize Git and Push

```
# Initialize git repository  
git init  
  
# Add all files  
git add .  
  
# Create first commit  
git commit -m "Initial commit: E-Commerce Sales Dashboard"  
  
# Go to GitHub and create a new repository (e.g. ecommerce-sales-dashboard)  
  
# Link your local repo to GitHub (replace YOUR-USERNAME)  
git remote add origin https://github.com/YOUR-USERNAME/ecommerce-sales-dashboard.git  
  
# Push to GitHub  
git branch -M main  
git push -u origin main
```

Step 3: What Gets Uploaded

These files will be pushed to GitHub (node_modules and .env excluded if using .gitignore):

- All source code in `src/`
- `package.json` and `package-lock.json`
- `index.html`, `vite.config.ts`, `tsconfig.json`
- SQL migration files in `supabase/migrations/`
- `.gitkeep`

Step 4: For Anyone Cloning the Repo

```
# Clone the repo
git clone https://github.com/YOUR-USERNAME/ecommerce-sales-dashboard.git

# Install dependencies
cd ecommerce-sales-dashboard
npm install

# Create .env file with your Supabase credentials
echo "VITE_SUPABASE_URL=https://cykbfaejuizbfbxirr.supabase.co" > .env
echo "VITE_SUPABASE_ANON_KEY=YOUR_ANON_KEY" >> .env

# Start the dev server
npm run dev
```


11. Quick Reference Summary

Item	Value
Project name	ecommerce-sales-dashboard
Supabase project ID	cykbfaejuizbfbxirr
Supabase URL	https://cykbfaejuizbfbxirr.supabase.co
Tables	5 (categories, products, customers, orders, order_items)
Total rows	10,380 (6 + 24 + 200 + 3,383 + 6,767)
Migrations applied	3
Edge functions	0
Auth	None (public demo)
RLS	Enabled on all tables, SELECT-only policies
Frontend port	3000
Charts	4 (Area, Bar, Pie, Bar)
KPI cards	5
Theme	Dark navy
Font	Inter